To: May 29, 2026

Dr. Zhonghua Zheng
Editor, *Atmospheric Chemistry and Physics*

Dear Dr. Zheng,

We would like to submit the enclosed manuscript **"Physics-informed spatio-temporal graph neural networks for fine-resolution multi-variable urban weather nowcasting from sparse station observations"** by Zhang Q, Song J, and Liu H for possible publication in the journal of *Atmospheric Chemistry and Physics*.

Motivation. Sub-kilometer urban microclimates govern heat stress, building energy loads, and public health outcomes in cities that house over half the world's population. Accurate and timely urban weather nowcasting (1–6 h lead time) from sparse surface stations is therefore essential, yet remains technically challenging. Conventional numerical weather prediction at such fine resolutions is computationally prohibitive for real-time operational use, while purely data-driven models trained on isolated stations neglect the spatial coupling that governs urban boundary-layer processes. Our work addresses this gap by coupling physically motivated graph topologies with deep learning to enable station-to-grid nowcasting, a direction well aligned with ACP's interest in urban atmospheric processes and climate–health interactions.

Key contributions.

1. *Five physically motivated sub-graphs* encode proximity, thermal similarity, wind advection, terrain, and Local Climate Zone morphology, letting the model recover boundary-layer diagnostics—diurnal convective transitions, persistence decay, and the logarithmic wind–UHI coupling—directly from observations, without prescribed physical equations.
2. *Station-to-grid downscaling* turns 57 irregularly sited stations into 333 m gridded fields (140×140), retaining 92.7 % of observed spatial heterogeneity and reducing systematic WRF temperature bias by 67 %, humidity bias by 65 %, and wind speed bias by 25 %.
3. *Two-stage WRF-to-observation transfer learning* with Clausius–Clapeyron-motivated thermodynamic constraints mitigates the data sparsity challenge (1,416 training samples over 75 summer days) while keeping predictions physically consistent.

Broader significance. The framework yields station-specific thermal comfort indices (Apparent Temperature, WBGT, Heat Index, Wet-bulb Temperature) with 78–83 % error reductions during extreme heat episodes—directly useful for urban heat early-warning services.

Data and code. All source code and trained checkpoints are publicly archived on Zenodo (`https://doi.org/10.5281/zenodo.20436897`).

Disclosure. This manuscript has not been published or submitted elsewhere. All authors approved the submission and declare no competing interests. AI-assisted tools were used solely for language polishing; the authors bear full responsibility for the scientific content.

We look forward to your evaluation of this work. Should you require any additional information or revisions, we would be happy to respond promptly.

Sincerely yours,

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